Aiqon assignment

Predicting shipment ETA and delay propagation

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1. Summary

The system predicts the final delivery time of a shipment, the probability that it arrives more than 24 hours late, and which stage of the journey the delay comes from. The prediction is refreshed at three checkpoints: departure, port arrival and customs cleared. The main idea is that a delay already seen is carried forward, so the model only has to learn what the rest of the journey adds or recovers. On a time-ordered test set the mean absolute error (MAE) of the ETA goes from 22.5 h for the published schedule to 15.1 h at departure, 13.1 h at port arrival and 3.6 h after customs. All data is synthetic, so these numbers show that the method recovers the structure I put into the simulator. They are not an estimate of accuracy on real shipments.

2. Data and ground truth

I started with the Kaggle logistics and supply chain dataset, but as far as I could tell from its columns it gives one ETA variation value per shipment instead of stage timestamps, and no model beat a median baseline on it. That cannot show how a delay moves from one stage to the next, so I simulated 300 shipments during 2024 on 8 routes (for example Shanghai to Los Angeles and Chennai to Singapore). Every shipment goes through departure, sea transit, port handling, customs and trucking, each with a scheduled and an actual time. Delays are introduced like this:

- Departure delay is gamma distributed and grows with origin congestion and bad origin weather.
- At sea the ship recovers up to 30% of the departure delay by speeding up (capped at 4% of the sea time). Storms add delay.
- Berth waiting grows with the square of destination congestion. A ship more than 12 hours late loses its berth slot. Handling time depends on congestion and container count together.
- Customs adds time for inspections (more likely for hazardous cargo and in busy ports) and for a backlog when the port stage ran long. On routes where customs closes at weekends, a clearance that lands on Saturday or Sunday waits until Monday 09:00.
- Trucks do not start after 17:00, and destination weather slows trucking.
- To mimic tracking gaps, 7% of port arrival, 6% of port release and 10% of customs clearance timestamps are blanked.

Ground truth. The delay at a milestone is actual minus scheduled time in hours. A stage delay is the change between two milestones, and the five stage delays add up exactly to the final delay (checked in the code). The final delay is actual minus scheduled delivery, and a shipment is delayed if it is more than 24 hours late (46% of shipments; mean delay 25 h). The predicted ETA is the scheduled delivery plus the predicted final delay. The 24 hour limit is my own choice.

3. Method

Checkpoints and remaining delay. A prediction is made at departure, at port arrival and after customs, using only what is known at that moment. At each checkpoint I take the latest known cumulative delay (if a timestamp is missing, the one before it) and train the model on the remaining delay, which is the final delay minus the carried delay. The prediction is carried delay plus predicted remaining delay. This is how I model propagation: what is already late stays late unless later stages absorb it, and the model learns how much the later stages add.

Features. Stage delays, route, cargo, container count, planned stage hours, congestion and weather indices. To connect stages I added arrival delay x destination congestion, congestion x planned port hours, congestion x container count and a lost-slot flag. I also add calendar features for the carried-forward clearance time (scheduled clearance plus carried delay): weekday, hour, whether it falls on a weekend at a port that closes, and whether it is after the 17:00 trucking cut-off. Missing values are filled with the training median and a missing-indicator column. Models: ridge and gradient boosting for the remaining delay, and logistic regression and gradient boosting for delayed or not.

4. Evaluation

Shipments are in time order. The first 225 are used for training and the last 75 for testing. As a second check I use a 5-fold expanding-window time series split over all 300 shipments. I compare against three rules: schedule only (zero delay), route median delay, and carry forward (scheduled ETA plus the delay seen so far). Metrics are MAE and RMSE in hours for the ETA, and AUC, F1, precision and recall (threshold 0.5) for delayed or not.

5. Results

MAE (hours)	Test set			5-fold time series split		
	departure	**port arrival**	**customs**	**departure**	**port arrival**	**customs**
Schedule only	22.5	22.5	22.5	27.5 +/- 3.3	27.5 +/- 3.3	27.5 +/- 3.3
Route median	17.1	17.1	17.1	-	-	-
Carry forward	19.8	17.7	4.9	23.4 +/- 2.6	21.7 +/- 2.7	5.9 +/- 0.5
Ridge	15.5	13.5	4.1	17.9 +/- 2.0	15.3 +/- 1.5	5.5 +/- 1.4
Gradient boosting	**15.1**	**13.1**	**3.6**	**17.2 +/- 2.0**	**14.2 +/- 2.1**	**5.0 +/- 0.9**

Table 1. ETA error of the final delivery time. Ridge and gradient boosting predict the remaining delay.

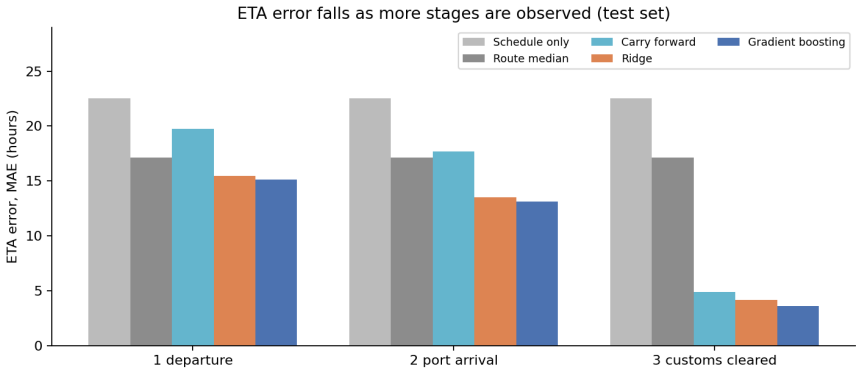


Figure 1. Test-set ETA error by checkpoint.

Checkpoint	Method	AUC	F1	Precision	Recall
1 departure	Carry forward	0.61	0.00	0.00	0.00
1 departure	Gradient boosting	0.78	0.68	0.70	0.66
2 port arrival	Carry forward	0.63	0.00	0.00	0.00
2 port arrival	Gradient boosting	0.80	0.70	0.68	0.72
3 customs cleared	Carry forward	0.97	0.89	0.96	0.83
3 customs cleared	Gradient boosting	0.95	0.90	0.90	0.90

Table 2. Delayed (more than 24 h late) or not, test set. The carry-forward rule never fired before customs, so its F1 is 0 there. Logistic regression gave similar numbers (in the notebook).

- After customs the error is a few hours. Even carrying the delay forward gives 4.9 h, and boosting improves this to 3.6 h, mostly by learning the 17:00 trucking cut-off.
- At departure and port arrival boosting cuts the schedule-only error by about a third and about 40%, but 13 to 15 hours of error remain. In this data the biggest random effects (berth wait, inspection, weekend hold) happen after arrival, so they cannot be seen earlier.
- Ridge and boosting are close. With 75 test shipments and 225 training rows I would not claim that one is better.
- Missing timestamps matter most at the last checkpoint: MAE 8.7 h with a gap against 3.9 h without (out-of-fold predictions). Using the latest known delay stops the model from breaking, but it cannot recover the lost information.
- The two short routes (Chennai-Singapore, Mundra-Jebel Ali) have the lowest errors at the first two checkpoints. At the last one, Nhava Sheva-Rotterdam and Shanghai-Los Angeles stand out at about 7 h, but with about 35 shipments per route I would not read much into it.
- Feature check: the calendar timing features reduce the error by 0.9 h at departure and 1.4 h at port arrival. The interaction features add at most 0.2 h, which is within noise, so this data does not show that they help.

6. Explaining the predicted delay

For the boosting models I compute SHAP values and add them up by stage group, so the explanation reads "port arrival and congestion added about X hours" instead of a list of columns. An explanation has two parts that add up exactly to the predicted delay: the delay already incurred in each completed stage (taken from the timestamps), and the delay still expected (a typical average plus a contribution per stage group). Figure 2 shows how the drivers change by checkpoint. At departure the largest groups are clearance timing and route and cargo; at port arrival it is port arrival and congestion (6.5 h mean absolute SHAP), followed by clearance timing; after customs the 17:00 cut-off dominates.

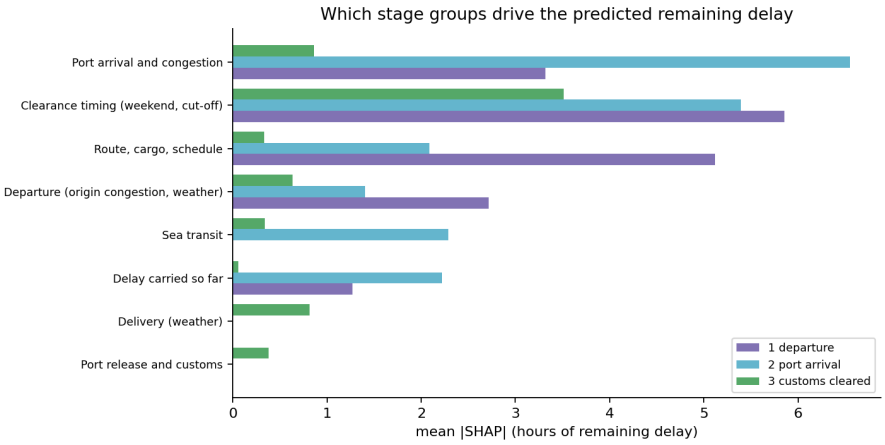


Figure 2. Mean absolute SHAP value by stage group on the test set.

Figure 3 follows one shipment, SH0251 (Colombo to Felixstowe, the largest delay in the test set). Its port arrival timestamp was missing, so at checkpoint 2 the model only had the departure delay and predicted about 13 hours late. The shipment was 85 hours late: the cargo was inspected, released on a Thursday afternoon, and clearance ran into the weekend until Monday morning. At checkpoint 3 the explanation shows 69 hours incurred in customs and 17 in port handling, and the predicted ETA is about 6 hours off. It is a useful reminder that a stage-based system can only explain delays after the stage has been observed.

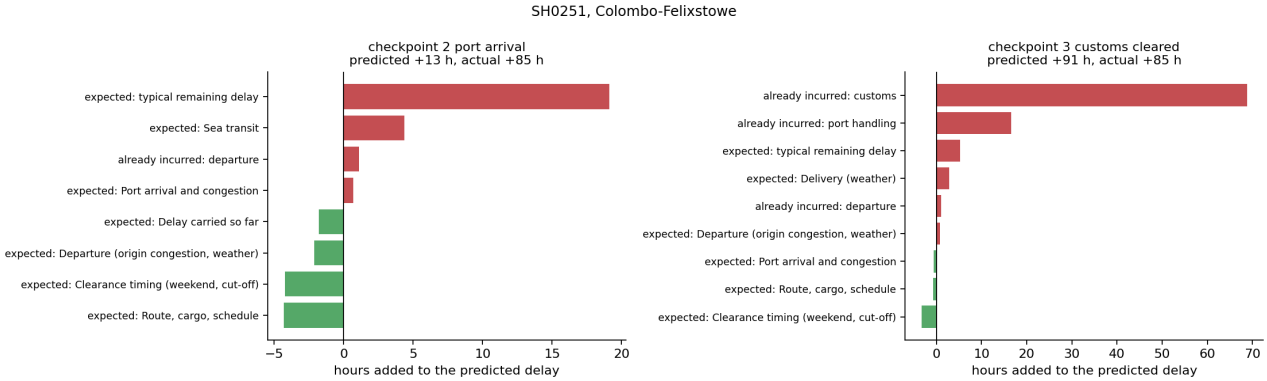


Figure 3. Explanation of one shipment at two checkpoints.

7. Assumptions and limitations

- All data is synthetic. Weekend closures, the 17:00 cut-off and the lost berth slot are rules I wrote and then built features to find, so the results are partly built in. Real data would contain effects I did not think of.
- The ETA is evaluated for final delivery only. Each prediction uses only information available at its checkpoint, and destination congestion at departure is a noisy forecast.
- Timestamps are missing at random. In real data, gaps are probably related to the delays themselves.
- 300 shipments and a 75-row test set give wide error bars. The test period also has fewer delays than the training period (39% against 48%) because congestion peaks in mid-year.
- SHAP describes the model, not the cause. Here I know the mechanism, but on real data it would be a starting point for investigation.
- There are no prediction intervals. A real ETA service should give a range, for example with quantile regression.
- Next steps: real port call or AIS data, quantile predictions, and a stage-by-stage model to compare with the remaining-delay approach.